Geometric sequences

1

a i Geometric ii $r = -9$

b i Arithmetic ii $d = 4$

2

a Geometric sequence

b Geometric sequence

c Not a geometric sequence

d Not a geometric sequence

3 Yes

4

a $\frac{u_2}{u_1} = 2$

b $\frac{u_3}{u_2} = 2$

c $\frac{u_4}{u_3} = 2$

d $u_5 = -128$

5 By choosing any term after the first and dividing it by the previous term.

6 $\frac{1}{4}$

7 $4 - \sqrt{15}$

8

a $6, 24, 96, 384$

b $7, -14, 28, -56$

9

a $108, 324$

b $-768, 3072$

c $\frac{27}{64}, \frac{81}{256}$

d $\frac{81}{4}, -\frac{243}{8}$

10 $n^5 - 5n^4$ and $n^6 + 5n^5$

11

a i $3, 12, 48, 192$ ii $r = 4$

b i $-4, 12, -36, 108$ ii $r = -3$

12

a $-5, 20, -80, 320, -1280$

b $3, -\frac{3}{5}, \frac{3}{25}, -\frac{3}{125}, \frac{3}{625}$

c $18, 12, 8, \frac{16}{3}, \frac{32}{9}$

Technology

13

a $u_1 r^5 = 557$

b $u_1 r^{12} = 255\,642$

c $u_1 = 7$ and $r = 2.4$

d $u_8 = 3211$

14

a $u_1 r^6 = 353$

b $u_1 r^{12} = 42$

c $u_1 = 2967$ and $r = 0.7$

d $u_9 = 171$

15

a $u_1 r^4 = 11$

b $u_1 r^{11} = 72$

c $u_1 = 4$ and $r = 1.3$

d $u_n = 4\left(1.3^{n-1}\right)$

16

a $u_1 r^3 = 33$

b $u_1 r^{13} = 952$

c $u_n = 12\left(1.4^{n-1}\right)$

17

a $x = 16$

b $r = \frac{2}{3}$

c $u_4 = 8$

Applications

18

a

Day	Mass of element D (g)
0	800
1	400
2	200
3	100
4	50

b Exponential

19

a 1.2

b 51.84

20

a $65 536

b $268 435 456

22

a

i 16

ii $11\frac{5}{9}$ kg

iii $3\frac{5}{9}$ minutes

iv $\frac{4}{9}$ miles

b

i 15

ii $\frac{5}{12}$ miles

c Beginner

23

a 12%

b $19\,994$

24

a $1:1$

b $20\,736\ \text{cm}^2$

c $n = 2$

25

a 1216 bacteria

b 9 bacteria